

Review Day

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

$$\lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h}$$

$$\lim_{t \rightarrow \infty} \frac{4t^2 - 3t + 2}{3t^3 + t^2 - t + 1}$$

Consider the function $f(x) = \frac{|x|}{x}$. Use a graph to explain why $f'(0)$ does not exist.

Sketch one function $f(x)$ that satisfies all of the following properties:

[2 pts] $f(0) = 2$

[2 pts] $\lim_{x \rightarrow -2^+} f(x) = \infty$

[2 pts] $\lim_{x \rightarrow -2^-} f(x) = -\infty$

[2 pts] $\lim_{x \rightarrow 4} f(x) = -2$

[2 pts] $\lim_{x \rightarrow -\infty} f(x) = 2$

Sketch the graph of one continuous function f that satisfies all of the following properties:

[2 pts] $f'(x) > 0$ for $x < -1$

[2 pts] $f'(x) = 0$ for $x = -1$

[2 pts] $f'(x) < 0$ for $x > -1$

[2 pts] $f''(x) < 0$ for $-2 < x < 1$

[2 pts] $f''(x) > 0$ for $x > 1$ and $x < -2$

Consider a money market fund, and let $f(t)$ be the balance, in dollars (\$), of the fund, where t is the number of years since 2021. Explain the meaning of each of the following statements. Include units in your answers.

$$f(0) = 24,263.00 \quad f(1) = 25,476.15 \quad f'(1) = 1,300.00 \quad f^{-1}(28,087.46) = 3 \quad (f^{-1})'(25,000) = 0.0007$$

Find Derivatives for each of the following functions:

$$y = 3t^5 - 5\sqrt{t} + \frac{7}{t}$$

$$y = e^x + 5^x$$

$$f(x) = \frac{3x^3 - 5x}{x^2}$$